

Positioning: structure versus pedagogy

Please use CSS style instead of skinparam padding

Author's note — not a literature survey.
These four families are derived from this system's own capabilities and dependency set, not from a citation search. Populate each family with the specific works you cite in Chapter 2, and check that the boundaries still hold once you do.

Learning platforms / LMS

courses, enrolment, grading

Rich *pedagogy*: assignments, progress, analytics, spaced review. Assumes content is **already** authored and structured. Ingestion is a manual upload.

Retrieval-augmented QA

grounded answers over a corpus

Answers *questions* over documents. Optimises faithfulness and citation. Does not durably materialise reusable learning objects.

Document conversion / layout AI

PDF to markup or markdown

Recovers *structure*: pages, blocks, tables, reading order. Produces no assessable objects. No learner model.

//This system consumes tools from this family: PyMuPDF for extraction, and MarkItDown / LlamaParse / MinerU / OpenDataLoader as swappable text front-ends.//

pedagogy,
but no ingestion

grounded answers,
but nothing persisted

structure,
but no pedagogy

THIS WORK

lecture materials to structured educational content

- ingests raw slide decks as the *primary* input
- materialises persistent, addressable objects: slides, MCQs with Bloom levels, embeddings
- layers pedagogy on the derived objects: grounded tutor, SM-2 review, mock exams
- keeps a human-in-the-loop correction path
- server-authoritative: the database defines what a slide *is* once parsed

The distinguishing move is **materialisation**. A RAG system answers a question and forgets; this pipeline writes durable objects that several independent features then consume — the same quiz_questions rows feed in-lecture quizzes, the SRS card factory (with no further LLM call) and mock-exam sampling. One extraction, many pedagogical uses.

Grayscale-safe: families are distinguished by position and label, not by fill.